Lab 6: Radiometer

Goal: Measure rotation speed at three lamp distances. Compare speed with measured lux and check whether it changes during a run.

A. Set the distances

1. Keep the radiometer upright and the camera fixed. Use the **same LED lamp at the same power** for every test. Aim the lamp at the rotor height.

2. Measure the horizontal gap from the **light-emitting front surface of the lamp to the nearest outer glass surface of the bulb**, at rotor height. Use the distances below; do not measure from the lamp stand or bulb center.

Recording	Lamp-to-glass gap
First test	10 cm
Second test	15 cm
Third test	20 cm
Repeat after cooling	10 cm

3. Place the black background **25 cm behind the rear glass surface**, outside the strongest direct beam. Keep it and the camera fixed. At each lamp distance, measure lux beside the bulb at rotor height, with the meter facing the lamp. Remove the meter if it blocks the view.

These close distances apply to the course LED lamp. Keep the lamp and glass from touching; do not substitute a hot incandescent/halogen lamp at these gaps.

B. Camera settings

Resolution: 1280 x 1024 pixels.
Capture frame rate: 1000 frames/s. The remaining camera settings are preset at the station.

C. Check, then record

4. Record a short test. In Playback, check that a selected vane passage is clear. Watch one full turn and count **p = selected passage events per revolution**. If following one identifiable vane, p = 1; do not assume p = 4.

5. Record at least **seven consecutive passage marks** (six intervals). Use real recording time: at 1000 fps, 10,000 frames represent 10 seconds, regardless of slow-motion playback length. Check the available recording time before taking all four tests. If it cannot hold enough passages, extend the station recording duration before proceeding.

6. Start from a cooled bulb with the rotor at rest. Start recording, switch on the lamp, and capture the passages. Save the useful segment as **H.264, Text Overlay off**. Name the movies **10cm_1.mp4, 15cm.mp4, 20cm.mp4 and 10cm_repeat.mp4**. Note the light-on frame in Playback.

7. Let the bulb return to a comparable cooled state between tests. Move only the lamp to the next distance. Repeat 10 cm last. If a test starts warm, note that fact; do not call it a cooled repeat.

Keep for each movie: lamp distance, measured lux, capture fps, p and light-on frame. Keep the original movie.

Analyze the recordings

D. Mark passages and calculate rpm

8. Download and unzip **MATLAB files > Complete bundle** from the Lab 6 page. Run **Mark_Lab6_Passage_Frames.m** and select a movie. Left/Right navigate; Up/Down change the step; Space marks; Backspace undoes; Q saves.

9. Use a one-frame step at each crossing. Mark the **same passage event in the same direction**, in increasing frame order. The program saves a file ending **_passages.csv**.

10. Run **Run_Lab6_Radiometer.m**, select the passage CSV, and enter **1000 fps** and your verified p. If the actual capture rate differs, enter that actual rate. Check unexpected intervals against the movie.

One-interval check: elapsed seconds = frame-number difference / capture fps.
rpm = 60 / (p x elapsed seconds). Seven marks give six intervals.

E. Compare the three distances

11. Open **Lab6_Comparison_Log.csv**. The four filenames and distances are already entered. Fill the measured lux, p, light-on frame and initial state. Keep capture_fps = 1000 unless the actual camera rate differed.

12. Choose the **same physical start/end times after light-on** that all four movies cover, with enough passage marks. Enter those times in window_start_s and window_end_s for every row. Time after light-on = (frame - light-on frame) / capture fps.

13. Run **Compare_Lab6_Illumination_Runs.m** and select the completed log. Review rpm versus time and mean rpm versus measured lux. Compare the first and repeated 10 cm tests, and early versus late speed. Results use the first-to-last accepted passage, not unrecorded time.

F. Put these results in the report

1. One labeled setup photo and a small table of distance, lux and mean rpm.
2. Cumulative turns and rpm versus time for one recording.
3. Mean rpm versus lux, with the repeated 10 cm test identified.
4. A short explanation of early/late changes and repeatability, using your Q6 molecular Monte Carlo result to discuss temperature contrast and gas momentum transfer.

Use only relevant theory. Lux is not absorbed radiant power, and the simple molecular wall model does not directly predict the bulb rpm.

Before leaving

Check that all four movies, passage CSVs, the completed comparison log and MATLAB figures are saved and shared with your group. Complete Q6 in Pre-Lab 6 before your section. **Lab 6 report: December 7, 11:59 PM.**